Contents

Chapter 1

Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation

1.1 Bilayer extended Hubbard model: Impurity model

[1] We approach the heavy-fermion problem by starting from a bilayer extended Hubbard model, consisting of two layers (f and c). Towards studying this lattice model, we adopt a two-layer impurity problem that hosts a correlated impurity site in each layer (S_f and S_d):

$$H_{\text{aux}} = H_{1\text{p}} + H_f + H_d + H_{fd} \ , \tag{1.1}$$

where $H_{1\text{p}}$ is the Hamiltonian for the non-interacting itinerant electrons of either layer,

$$H_{1\text{p}} = -t \sum_{\sigma,\alpha} \sum_{\langle i,j \rangle} \left(c^\dagger_{i,\sigma,\alpha} c_{j,\sigma,\alpha} + \text{h.c.} \right) - \sum_{i,\sigma,\alpha} \mu_\alpha n_{i,\sigma,\alpha} \ , \tag{1.2}$$

such that α sums over the two layers f and d. H_f and H_d describe the dynamics of the correlated impurity sites (and their local neighbourhood) in each layer:

$$\begin{aligned} H_f &= -\frac{U_f}{2}(n_{f,\uparrow} - n_{f,\downarrow})^2 + \sum_{Z \in \text{NN}} \Bigg[-V \sum_\sigma \left(c^\dagger_{f,\sigma} c_{Z,\sigma,f} + \text{h.c.} \right) + \frac{J}{4} \sum_{\alpha,\beta} \mathbf{S}_f \cdot \frac{1}{2} \boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\alpha,f} c_{Z,\beta,f} \\ &\qquad - \frac{W_f}{4} \left(n_{Z,\uparrow,f} - n_{Z,\downarrow,f} \right)^2 \Bigg] \ , \\ H_d &= -\eta_d n_d - \frac{U_d}{2}(n_{d,\uparrow} - n_{d,\downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_\sigma \left(c^\dagger_{d,\sigma} c_{Z,\sigma,d} + \text{h.c.} \right) + \frac{J}{4} \sum_{\alpha,\beta} \mathbf{S}_d \cdot \frac{1}{2} \boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\alpha,d} c_{Z,\beta,d} \right] \ , \end{aligned} \tag{1.3}$$

where $c^\dagger_{\nu,\sigma,\alpha}$ can refer to creation operator for either the $f-$layer. Finally, H_{fd} represents the inter-layer hybridisation:

$$H_{fc} = -V_\perp \sum_\sigma \left(c^\dagger_{d,\sigma} c_{f,\sigma} + \text{h.c.} \right) \ . \tag{1.4}$$

1.2 Tiling Into Bilayer Extended Hubbard Model

Tiling the impurity model leads to a bilayer extended Hubbard model on the 2D square lattice. In order to tile our auxiliary model to restore translation invariance, we identify the cluster which acts as the "unit" cell of translation. As discussed in the text surrounding eq. **??**, a good choice for the unit cell is the impurity site and it's surrounding correlated environment. Placing the impurity sites at $\boldsymbol{r}_d$ in each layer, we get

$$H_{\text{clust}}(\boldsymbol{r}_d) = H_f(\boldsymbol{r}_d) + H_d(\boldsymbol{r}_d) + H_{fc}(\boldsymbol{r}_d) \ . \tag{2.1}$$

The details of tiling a cluster Hamiltonian were already shown in Sec. **??**; the general idea is to use manybody translation operators (details in Sec. **??**) to center the cluster Hamiltonian at various sites of the lattice, such that combining all such terms leads to a periodised Hamiltonian. we write down the final result here:

$$\begin{aligned}
H_{\text{latt}} &= \sum_r T^\dagger(\boldsymbol{r}) H_{\text{clust}}(\boldsymbol{r}_d) T(\boldsymbol{r}) \\
&= -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(c^\dagger_{i,\sigma}c_{j,\sigma} + \text{h.c.}\right) - \sum_i\left[\eta_d n_{i,d} + \frac{U_d}{2}\left(n_{i,\uparrow,d} - n_{i,\downarrow,d}\right)^2\right] + \frac{2J}{Z}\sum_{\langle i,j\rangle}\boldsymbol{S}_{i,d}\cdot\boldsymbol{S}_{j,d} \\
&\quad - \frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(f^\dagger_{i,\sigma}f_{j,\sigma} + \text{h.c.}\right) - \frac{U_f - W_f}{2}\sum_i\left(n_{i,\uparrow,f} - n_{i,\downarrow,f}\right)^2 + \frac{2J}{Z}\sum_{\langle i,j\rangle}\boldsymbol{S}_{i,f}\cdot\boldsymbol{S}_{j,f} \\
&\quad - V_\perp\sum_{i,\sigma}(f^\dagger_{i,\sigma}c_{i,\sigma} + \text{h.c.})
\end{aligned} \tag{2.2}$$

Tiling the impurity model therefore leads to a bilayer extended Hubbard model, with effective couplings that depend on the auxiliary model couplings. The value of η_d defines the effective chemical potential of the $d-$layer; a non-zero value moves the layer away from half-filling. The $f-$layer is pinned at half-filling.

Following Sec. **??**, the retarded $k-$space Greens function in either layer can be expressed in terms of auxiliary model Greens functions:

$$G(\boldsymbol{k},\sigma,\alpha;\omega) = \sum_\nu C_{\nu,\boldsymbol{k}\alpha}\mathcal{G}(\mathcal{T}_{\nu,\sigma,\alpha};\omega - \mathcal{E}_\nu(\boldsymbol{k})) \tag{2.3}$$

where $\alpha = f, d$ denotes the layer in which the Greens function is being calculated, the modes ν indicate the operators which diagonalise the Hamiltonian $H_{1\text{p}}$, with energies $\mathcal{E}_\nu(\boldsymbol{k})$:

$$H_{1\text{p}} = -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(c^\dagger_{i,\sigma}c_{j,\sigma} + \text{h.c.} + f^\dagger_{i,\sigma}f_{j,\sigma} + \text{h.c.}\right) - V_\perp\sum_{i,\sigma}(f^\dagger_{i,\sigma}c_{i,\sigma} + \text{h.c.}) - \sum_i \eta_d n_{i,d} \ , \tag{2.4}$$

and the $\mathcal{T}-$matrices are defined in eq. **??**:

$$\mathcal{T}_{\nu,\sigma,\alpha} = \left[\left(\sum_{\sigma'} c^\dagger_{\boldsymbol{\alpha}\sigma} + \text{h.c.}\right) + \left(S^+_{\boldsymbol{\alpha}} + \text{h.c.}\right) + \left(C^+_{\boldsymbol{\alpha}} + \text{h.c.}\right)\right] c_{\nu,\sigma} \ , \tag{2.5}$$

In order to calculate $\mathcal{E}_\nu(\boldsymbol{k})$, we introduce the $k-$space modes via a Fourier transform:

$$c^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i} c^\dagger_{\boldsymbol{k},\sigma}~, \quad f^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i} f^\dagger_{\boldsymbol{k},\sigma}~. \tag{2.6}$$

In terms of these modes, the 1−particle Hamiltonian decouples in $k-$space:

$$H_{1\text{p}} = \sum_{\boldsymbol{k},\sigma}\left[(\epsilon_{\boldsymbol{k}} - \eta_d)(n_{\boldsymbol{k},\sigma,d} + n_{\boldsymbol{k},\sigma,f}) - V_\perp(c^\dagger_{\boldsymbol{k},\sigma} f_{\boldsymbol{k},\sigma} + \text{h.c.})\right]~. \tag{2.7}$$

The diagonal modes are given by

$$\nu^\dagger_{\boldsymbol{k},\sigma,\pm} = C_{\pm,k} c^\dagger_{\boldsymbol{k},\sigma} + \sqrt{1 - C^2_{\pm,k}} f^\dagger_{\boldsymbol{k},\sigma}~, \tag{2.8}$$

the coefficients obtained from diagonalising the matrix

$$H_{1\text{p}}(\boldsymbol{k}) = \begin{pmatrix} \epsilon_{\boldsymbol{k}} & -V_\perp \\ -V_\perp & \epsilon_{\boldsymbol{k}} - \eta_d \end{pmatrix}~. \tag{2.9}$$

The energies are given by $\mathcal{E}_\pm(\boldsymbol{k}) = \epsilon_k - \eta_d/2 \pm \Delta$, with ϵ_k being $-\frac{4V}{Z}(\cos k_x + \cos k_y)$ for a 2D square lattice and $\Delta = \sqrt{V_\perp^2 + \eta_d^2/4}$. Substituting into the expression for the lattice Greens function, we get

$$\begin{aligned} G(\boldsymbol{k}, d; \omega) =& C_{+,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k}+,d}, \mathcal{T}^\dagger_{\boldsymbol{k},d}; \omega - \mathcal{E}_+(\boldsymbol{k})) + C_{-,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k}-,d}, \mathcal{T}^\dagger_{\boldsymbol{k},d}; \omega - \mathcal{E}_-(\boldsymbol{k})) \\ &+ C^*_{+,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},d}, \mathcal{T}^\dagger_{\boldsymbol{k}+,d}; \omega - \mathcal{E}_+(\boldsymbol{k})) + C^*_{-,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},d}, \mathcal{T}^\dagger_{\boldsymbol{k}-,d}; \omega - \mathcal{E}_-(\boldsymbol{k}))~. \end{aligned} \tag{2.10}$$

Following eq. **??**, we can similarly write down an expression the local lattice Greens function for the $f-$ and $d-$electrons. For that, we define the local diagonal modes $\nu_\pm$ obtained from the local Hamiltonian

$$H_{\text{loc}} = \begin{pmatrix} 0 & -V_\perp \\ -V_\perp & 0 - \eta_d \end{pmatrix}~, \tag{2.11}$$

and their associated energies $\varepsilon_\pm$. Within the impurity model, these local bonding/antibonding modes can be constructed on the impurity sites ($\nu_\pm$) or on the neighbouring bath sites, ($\nu^Z_\pm$). The local Greens function can be expressed in terms of these modes:

$$\begin{aligned} G_{dd}(\omega) =& \frac{1}{2}\sum_\nu C_{\nu,d}\left[\mathcal{G}(\nu\sigma, d\sigma; \omega - \varepsilon_\nu) + \mathcal{G}(d\sigma, \nu\sigma; \omega - \varepsilon_\nu)\right] \\ &+ \frac{1}{2}\sum_{\nu^Z} C_{\nu^Z, Z_d}\left[\mathcal{G}(\mathcal{T}_{Z_d,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z}) + \mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z})\right]~. \end{aligned} \tag{2.12}$$

The presence of local hybridisation between the $d-$ and $f-$ orbitals makes the inter-orbital Greens function important as well. Following eq. **??**, we have

$$\begin{aligned} G_{fd}(\omega) =& \frac{1}{2}\sum_\nu\left[C_{\nu,f}\mathcal{G}(\nu\sigma, d\sigma; \omega - \varepsilon_\nu) + C_{\nu,d}\mathcal{G}(f\sigma, \nu\sigma; \omega - \varepsilon_\nu)\right] \\ &+ \frac{1}{2}\sum_{\nu^Z}\left[C_{\nu^Z, Z_f}\mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z}) + C_{\nu^Z, Z_d}\mathcal{G}(\mathcal{T}_{Z_f,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z})\right]~. \end{aligned} \tag{2.13}$$

Using these, we can construct the hybridised Greens function associated with the states ν:

$$G_{\nu_+}(\omega) = C^2_{+,d}G_{dd}(\omega) + C^2_{+,f}G_{ff}(\omega) + C_{+,d}C_{+,f}\left[G_{df}(\omega) + G_{fd}(\omega)\right] , \tag{2.14}$$

1.3 Unitary RG Analysis of The Bilayer Lattice-Embedded ES-IAM

In the limit of large U_α, we carry out a Schrieffer-Wolff transformation and work with the following low-energy Hamiltonian:

$$\begin{aligned}H_{\text{aux}} = &\sum_{\mathbf{k},\sigma,\alpha}\epsilon_{\mathbf{k},\alpha}n_{\mathbf{k},\sigma,\alpha} + \sum_{\alpha}\sum_{Z\in\text{NN}}\left[\frac{1}{4}J_\alpha\sum_{\sigma,\sigma'}\mathbf{S}_\alpha\cdot\frac{1}{2}\boldsymbol{\sigma}_{\sigma\sigma'}c^\dagger_{Z,\sigma,\alpha}c_{Z,\sigma',\alpha} - \frac{W_\alpha}{4}\left(n_{Z,\uparrow,\alpha} - n_{Z,\downarrow,\alpha}\right)^2\right]\\ &+ J_\perp\mathbf{S}_f\cdot\mathbf{S}_d\ .\end{aligned} \tag{3.1}$$

As we will see, additional terms are generated within the Hamiltonian as we integrate out high-energy degrees of freedom in the conduction baths of the two impurity models. These terms connect the impurity of the $d-(f-)$layer to the nearest-neighbour conduction bath sites of the $f-(d-)$layer, It is therefore necessary to start with an enlarged impurity model Hamiltonian that contains these couplings as well:

$$\begin{aligned}H_{\text{aux}} = &\sum_{\mathbf{k},\sigma,\alpha}\epsilon_{\mathbf{k},\alpha}n_{\mathbf{k},\sigma,\alpha} + \sum_{\alpha}\sum_{Z\in\text{NN}}\left[\frac{1}{4}\sum_{\sigma,\sigma'}\left(J_\alpha\mathbf{S}_\alpha + K_{\bar{\alpha}}\mathbf{S}_{\bar{\alpha}}\right)\cdot\frac{1}{2}\boldsymbol{\sigma}_{\sigma\sigma'}c^\dagger_{Z,\sigma,\alpha}c_{Z,\sigma',\alpha}\right.\\ &\left.-\frac{W_\alpha}{4}\left(n_{Z,\uparrow,\alpha} - n_{Z,\downarrow,\alpha}\right)^2\right] + J\mathbf{S}_f\cdot\mathbf{S}_d\ .\end{aligned} \tag{3.2}$$

Note the additional couplings $K_{\bar{\alpha}}$, where $\bar{\alpha}$ stands for the layer opposite to α.

The two Kondo couplings J and K and the bath interaction W acquire $k-$dependence:

$$\begin{aligned}J(\mathbf{k},\mathbf{q}) &= \frac{J}{2}\left[\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y)\right] ,\\ K(\mathbf{k},\mathbf{q}) &= \frac{K}{2}\left[\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y)\right] ,\\ W(\mathbf{k},\mathbf{q},\mathbf{k}',\mathbf{q}') &= \frac{W}{2}\left[\cos(\mathbf{k}_x - \mathbf{q}_x + \mathbf{k}'_x - \mathbf{q}'_x) + \cos(\mathbf{k}_y - \mathbf{q}_y + \mathbf{k}'_y - \mathbf{q}'_y)\right] .\end{aligned} \tag{3.3}$$

The particle-hole transformed state $\bar{\mathbf{q}}$ associated with $\mathbf{q}$ is defined as $\bar{\mathbf{q}} = \mathbf{q} + (\pi,\pi)$ if $\mathbf{q}$ is in the first or third quadrant and $\mathbf{q} + (\pi,-\pi)$ is in the second or fourth quadrant.

In order to study the low-energy physics of the impurity model, we iteratively integrate out high-energy degrees of freedom using the unitary RG method. Let the momentum states being decoupled from the two layers be $|\mathbf{q},\sigma,f\rangle$ and $|\mathbf{q},\sigma,d\rangle$ if they are above the Fermi energy (unoccupied in the low-energy configuration) and $|\bar{\mathbf{q}},\sigma,f\rangle$ and $|\bar{\mathbf{q}},\sigma,d\rangle$ if they are below the Fermi energy (occupied in the low-energy configuration) For every shell of conduction bath states of width ΔD integrated out, the Hamiltonian couplings renormalise by folding in the effects of virtual excitations to these states.

The full set of impurity-mediated scattering processes that lead to UV-IR mixing is:

Particle sector excitation These involve excited states above the Fermi energy:

$$\begin{aligned} P_{\text{dir}}(\alpha) &= \frac{1}{2}\sum_{k,q} J_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{q,\sigma_1,\alpha} c_{k,\sigma_2,\alpha} , \\ P_{\text{ind}}(\alpha) &= \frac{1}{2}\sum_{k,q} K_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{q,\sigma_1,\bar\alpha} c_{k,\sigma_2,\bar\alpha} . \end{aligned} \tag{3.4}$$

Hole sector These involve excited states below the Fermi energy:

$$\begin{aligned} H_{\text{dir}}(\alpha) &= \frac{1}{2}\sum_{k,q} J_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{k,\sigma_1,\alpha} c_{q,\sigma_2,\alpha} , \\ H_{\text{ind}}(\alpha) &= \frac{1}{2}\sum_{k,q} K_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{k,\sigma_1,\bar\alpha} c_{q,\sigma_2,\bar\alpha} . \end{aligned} \tag{3.5}$$

Mixed sector These involve simultaneous particle and hole excitations:

$$\begin{aligned} M_{\text{dir}}(\alpha) &= \frac{1}{2}\sum_{q,\bar q} J_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{q,\sigma_1,\alpha} c_{\bar q,\sigma_2,\alpha} , \\ M_{\text{ind}}(\alpha) &= \frac{1}{2}\sum_{q,\bar q} K_{\bar\alpha} \mathbf{S}_{\bar\alpha} \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{q,\sigma_1,\bar\alpha} c_{\bar q,\sigma_2,\bar\alpha} . \end{aligned} \tag{3.6}$$

1.3.1 Particle/hole scattering processes: direct with direct

We already have the renormalisation group equations for the couplings J_α in the case of $J = K = 0$, arising from P_{dir} and H_{dir}:

$$\frac{\Delta J_f(\mathbf{k}_1,\mathbf{k}_2)}{\Delta D} = -\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[J_f(\mathbf{k}_1,\mathbf{q})J_f(\mathbf{q},\mathbf{k}_2) + 4J_f(\mathbf{q},\bar{\mathbf{q}})W_f(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)\right] G^{(0)}_f(\mathbf{q}) , \tag{3.7}$$

where $\mathbf{q}$ sums over the UV states being integrated out, $\rho(\mathbf{q})$ is the density of states for the momentum states being integrated out and $G^{(0)}_f$ is the propagator for the excited intermediate state (for $J = 0$):

$$G^J_\alpha(\mathbf{q}) = \frac{1}{\omega - |\epsilon_\alpha(\mathbf{q})|/2 + J_\alpha(\mathbf{q})/4 + W_\alpha(\mathbf{q})/2} . \tag{3.8}$$

An identical equations holds for ΔJ_d, with all $f-$quantities replaced with $d-$quantities. Switching on J modifies the excitation energy in the propagator, while either of the Ks do not enter the denominator because either the impurity or the conduction bath labels do not match the ones fluctuating in this process:

$$G_\alpha(\mathbf{q}) = \frac{1}{1/G^J_\alpha - J\vec{S}_f\cdot\vec{S}_d} . \tag{3.9}$$

This expression can be simplified by noting that the operator in the denominator has only two eigenvalues, $-3/4$ in the singlet sector and $1/4$ in the triplet sector. Defining the projectors $\mathcal{P}_0$ and $\mathcal{P}_1$ for the two sectors, we have

$$\vec{S}_f \cdot \vec{S}_d = \frac{-3}{4}\mathcal{P}_0 + \frac{1}{4}\mathcal{P}_1, \quad \mathcal{P}_0 + \mathcal{P}_1 = \mathbb{I} \,. \tag{3.10}$$

Using these relations, we can write

$$\begin{aligned} G_f(\mathbf{q}) &= \frac{1}{1/G_f^J + 3J/4}\mathcal{P}_0 + \frac{1}{1/G_f^J - J/4}\mathcal{P}_1 \\ &= \left(\frac{1/4}{1/G_f^J + 3J/4} + \frac{3/4}{1/G_f^J - J/4}\right)\mathbb{I} + \left(\frac{1}{1/G_f^J + 3J/4} + \frac{1}{1/G_f^J - J/4}\right)\vec{S}_f \cdot \vec{S}_d \,. \end{aligned} \tag{3.11}$$

Only the first operator renormalises the Kondo interaction:

$$\begin{aligned} \frac{\Delta J_\alpha}{\Delta D} = -\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[J_\alpha(\mathbf{k}_1, \mathbf{q})J_\alpha(\mathbf{q}, \mathbf{k}_2) + 4J_\alpha(\mathbf{q}, \bar{\mathbf{q}})W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)\right] \Bigg(&\frac{1}{4}\frac{1}{1/G_\alpha^J(\mathbf{q}) + 3J/4} \\ &+ \frac{3}{4}\frac{1}{1/G_\alpha^J(\mathbf{q}) - J/4}\Bigg) \,. \end{aligned} \tag{3.12}$$

We now simplify the second term, in conjunction with the two kinds of processes on either side: J^2 processes and JW processes. We first focus on the J^2 processes.

Effect of inter-layer term $\vec{S}_f \cdot \vec{S}_d$ on J^2 processes

Taking into account the other spin operators on either side of $G_f(\mathbf{q})$ in the full scattering process, we have

$$\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\beta}\sum_{\gamma,\delta} S_f^a \sigma_{\alpha,\beta}^a S_f^b S_d^b S_f^c \sigma_{\gamma,\delta}^c \left[\delta_{\beta,\gamma} c_{k,\alpha,f}^\dagger c_{q,\beta,f} c_{q,\gamma,f}^\dagger c_{k',\delta,f} + \delta_{\alpha,\delta} c_{\bar{q},\alpha,f}^\dagger c_{k',\beta,f} c_{k,\gamma,f}^\dagger c_{\bar{q},\delta,f}\right] \,. \tag{3.13}$$

Integrating out the UV states using $n_{\bar{q}\beta} = 1 - n_{q\beta} = 1$, we get

$$\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\beta,\gamma} S_f^a S_f^b S_d^b S_f^c \left[\sigma_{\alpha,\beta}^a \sigma_{\beta,\gamma}^c c_{k,\alpha,f}^\dagger c_{k',\gamma,f} + \sigma_{\alpha,\beta}^a \sigma_{\gamma,\alpha}^c c_{k',\beta,f} c_{k,\gamma,f}^\dagger\right] \,. \tag{3.14}$$

Using $\sum_\beta \sigma_{\alpha,\beta}^a \sigma_{\beta,\gamma}^c = (\sigma^a \sigma^c)_{\alpha,\gamma} = \delta_{ac}\delta_{\alpha\gamma} + i\epsilon_{acm}\sigma_{\alpha,\gamma}^m$ in the first term and $\sum_\alpha \sigma_{\gamma,\alpha}^c \sigma_{\alpha,\beta}^a = (\sigma^c \sigma^a)_{\gamma,\beta} = \delta_{ca}\delta_{\gamma\beta} - i\epsilon_{acm}\sigma_{\gamma,\beta}^m$ in the second term, relabelling $\beta \to \gamma$ and $\gamma \to \alpha$ there and using $c_{k',\gamma} c_{k,\alpha}^\dagger = \delta_{k,k'}\delta_{\alpha,\gamma} - c_{k,\alpha}^\dagger c_{k',\gamma}$, we get

$$\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\gamma} S_f^a S_f^b S_d^b S_f^c \left[\delta_{a,c}\delta_{k,k'}\delta_{\alpha,\gamma} + 2i\epsilon_{acm}\sigma_{\alpha,\gamma}^m c_{k,\alpha,f}^\dagger c_{k',\gamma,f}\right] \,. \tag{3.15}$$

We also use the spin-half identity

$$\begin{aligned}\sigma^a\sigma^b\sigma^c =&(\delta_{ab} + i\epsilon_{abp}\sigma^p)\sigma^c \\ &= \delta_{ab}\sigma^c + i\epsilon_{abp}(\delta_{pc} + i\epsilon_{pcr}\sigma^r) \\ &= \delta_{ab}\sigma^c - \delta_{ac}\sigma^b + \delta_{bc}\sigma^a + i\epsilon_{abc}\ ,\end{aligned} \tag{3.16}$$

which when contracted with ϵ_{acm} gives

$$\begin{aligned}\sum_{a,c}\epsilon_{acm}\sigma^a\sigma^b\sigma^c &= \sum_{a,c}\epsilon_{acm}(\delta_{ab}\sigma^c + \delta_{bc}\sigma^a + i\epsilon_{abc}) \\ &= \sum_{a,c}(\epsilon_{acm}\delta_{ab}\sigma^c + \epsilon_{cam}\delta_{ba}\sigma^c - i\epsilon_{acm}\epsilon_{acb}) \\ &= -2i\delta_{bm}\ .\end{aligned} \tag{3.17}$$

For the first term of eq. 3.15, we get

$$\begin{aligned}\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\gamma}S_f^aS_f^bS_d^bS_f^c\delta_{a,c}\delta_{k,k'}\delta_{\alpha,\gamma} &= \delta_{k,k'}\sum_{a,b}S_f^aS_f^bS_f^aS_d^b \\ &= \frac{1}{4}\delta_{k,k'}\sum_b(S_f^b - 3S_f^b + S_f^b)S_d^b \\ &= -\frac{1}{4}\delta_{k,k'}\sum_b S_f^bS_d^b\ .\end{aligned} \tag{3.18}$$

For the second term, we obtain

$$\begin{aligned}\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\gamma}S_f^aS_f^bS_d^bS_f^c2i\epsilon_{acm}\sigma^m_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f} &= \frac{1}{4}\sum_b\sum_{\alpha,\gamma}\delta_{bm}S_d^b\sigma^m_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f} \\ &= \frac{1}{2}\sum_b S_d^b\frac{1}{2}\sum_{\alpha,\gamma}\sigma^b_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f}\end{aligned} \tag{3.19}$$

Substituting these in the expression for the renormalisation gives

$$\begin{aligned}&\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\gamma}S_f^aS_f^bS_d^bS_f^c\left[\delta_{a,c}\delta_{k,k'}\delta_{\alpha,\gamma} + 2i\epsilon_{acm}\sigma^m_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f}\right] \\ &= -\frac{1}{4}\mathbf{S}_f\cdot\mathbf{S}_d\delta_{k,k'} + \frac{1}{2}\sum_{\alpha,\gamma}\mathbf{S}_d\frac{1}{2}\sigma_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f}\ .\end{aligned} \tag{3.20}$$

The first term leads to the renormalisation of the inter-layer hybridisation $J_\perp$:

$$\frac{\Delta J_\perp}{\Delta D} = -\frac{1}{4}\int_{UV}d\mathbf{q}\,\rho(\mathbf{q})\int_{IR}d\mathbf{k}\rho(\mathbf{k})\sum_\alpha J_\alpha(\mathbf{k},\mathbf{q})^2\left(\frac{1}{1/G^J_\alpha + 3J/4} + \frac{1}{1/G^J_\alpha - J/4}\right)\ , \tag{3.21}$$

while the second term gives rise to the indirect impurity-bath Kondo couplings K_α mentioned earlier above eq. 3.2:

$$\frac{\Delta K_\alpha}{\Delta D} = \frac{1}{2}\int_{UV}d\mathbf{q}\,\rho(\mathbf{q})J_{\bar{\alpha}}(\mathbf{k}_1,\mathbf{q})J_{\bar{\alpha}}(\mathbf{q},\mathbf{k}_2)\left(\frac{1}{1/G^J_{\bar{\alpha}} + 3J/4} + \frac{1}{1/G^J_{\bar{\alpha}} - J/4}\right)\ , \tag{3.22}$$

Effect of inter-layer term $\vec{S}_f \cdot \vec{S}_d$ on JW processes

We now turn to the effect of the $\vec{S}_f \cdot \vec{S}_d$ in eq. 3.11 on processes that involve a Kondo scattering J and a bath scattering W. The non-vanishing processes are of the form

$$\begin{aligned} P_1 &= (S_f^+ \mathbf{S_f} \cdot \mathbf{S_d} + \mathbf{S_f} \cdot \mathbf{S_d} S_f^+) c^\dagger_{k,\downarrow} c_{k',\uparrow}, \\ P_2 &= P_1^\dagger, \\ P_z &= (S_f^z \mathbf{S_f} \cdot \mathbf{S_d} + \mathbf{S_f} \cdot \mathbf{S_d} S_f^z) \sum_\sigma \sigma c^\dagger_{k,\sigma} c_{k',\sigma} \ . \end{aligned} \tag{3.23}$$

In order to evaluate them, we note that for any $a \in x, y, z$, we have

$$S_1^a \mathbf{S_1} \cdot \mathbf{S_2} + \mathbf{S_1} \cdot \mathbf{S_2} S_1^a = \sum_b \{S_1^a, S_1^b\} S_2^b = \frac{1}{4} \sum_b 2\delta_{a,b} S_2^b = \frac{1}{2} S_2^a \ . \tag{3.24}$$

Using this, we obtain

$$\begin{aligned} P_1 &= \frac{1}{2} S_d^+ c^\dagger_{k,\downarrow} c_{k',\uparrow}, \\ P_2 &= \frac{1}{2} S_d^- c^\dagger_{k,\uparrow} c_{k',\downarrow}, \\ P_z &= \frac{1}{2} S_d^z \sum_\sigma \sigma c^\dagger_{k,\sigma} c_{k',\sigma} \ . \end{aligned} \tag{3.25}$$

Since the bath sites are in the $f-$layer, these terms again lead to the renormalisation of the indirect couplings K_α. Combining with eq. 3.22, we get

$$\frac{\Delta K_\alpha}{\Delta D} = \frac{1}{2} \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[J_{\bar{\alpha}}(\mathbf{k}_1, \mathbf{q}) J_{\bar{\alpha}}(\mathbf{q}, \mathbf{k}_2) - 4 J_{\bar{\alpha}}(\mathbf{q}, \bar{\mathbf{q}}) W_{\bar{\alpha}}(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2) \right] \left(\frac{1}{1/G^J_{\bar{\alpha}} + 3J/4} + \frac{1}{1/G^J_{\bar{\alpha}} - J/4} \right) , \tag{3.26}$$

1.3.2 Particle/hole scattering processes: indirect with indirect

The processes that apply to this class are very similar to those participating in the direct-direct class; the only difference is that the bath degrees of freedom have to be transferred from $\alpha \to \bar{\alpha}$, due to the inter-layer nature of this coupling. As a result, we will have straightforward counterparts to the renormalisation equations obtained in the previous section, eqs. 3.12, 3.21 and 3.26. The first equation is for the renormalisation of the indirect coupling through itself,

$$\frac{\Delta K_\alpha}{\Delta D} = - \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[K_\alpha(\mathbf{k}_1, \mathbf{q}) K_\alpha(\mathbf{q}, \mathbf{k}_2) + 4 K_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_{\bar{\alpha}}(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2) \right] \left(\frac{1}{4} \frac{1}{1/G^K_\alpha(\mathbf{q}) + 3J/4} + \frac{3}{4} \frac{1}{1/G^K_\alpha(\mathbf{q}) - J/4} \right) . \tag{3.27}$$

where the indirect Greens function G^K_α is defined as

$$G^K_\alpha(\mathbf{q}) = \frac{1}{\omega - |\epsilon_{\bar{\alpha}}(\mathbf{q})|/2 + K_\alpha(\mathbf{q})/4 + W_{\bar{\alpha}}(\mathbf{q})/2}~, \tag{3.28}$$

and reflects the fact the excitations are happening on impurity and bath states in opposite layers. The second equation is for the effect of the inter-layer term through the denominator of such processes, leading to a renormalisation of the inter-layer itself:

$$\frac{\Delta J_\perp}{\Delta D} = -\frac{1}{4}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q}) \int_{IR} d\mathbf{k}\rho(\mathbf{k}) \sum_\alpha K_\alpha(\mathbf{k},\mathbf{q})^2 \left(\frac{1}{1/G^K_{\bar{\alpha}} + 3J/4} + \frac{1}{1/G^K_{\bar{\alpha}} - J/4}\right)~. \tag{3.29}$$

The final equation is for the renormalisation of the direct Kondo coupling arising from indirect processes. Similar to how the direct Kondo coupling renormalised the indirect coupling, this occurs through two routes. The complete equation is

$$\frac{\Delta J_\alpha}{\Delta D} = \frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q}) \left[K_{\bar{\alpha}}(\mathbf{k}_1,\mathbf{q})K_{\bar{\alpha}}(\mathbf{q},\mathbf{k}_2) - 4K_{\bar{\alpha}}(\mathbf{q},\bar{\mathbf{q}})W_\alpha(\mathbf{k}_1,\mathbf{q},\mathbf{q},\mathbf{k}_2)\right] \left(\frac{1}{1/G^K_\alpha + 3J/4} + \frac{1}{1/G^K_\alpha - J/4}\right)~. \tag{3.30}$$

1.3.3 Particle/hole scattering processes: direct with indirect

These processes are composed of a direct Kondo excitation process followed by an indirect Kondo de-excitation process, or vice versa. They can be of the particle excitation or the hole excitation kind. Within the particle or hole sector, we again have two sets of processes depending on whether the initial process is direct or indirect:

a. $P_{\text{dir}}(d)\ G_d\ H_{\text{ind}}(f) + \text{h.c.}$

b. $P_{\text{dir}}(f)\ G_f\ H_{\text{ind}}(d) + \text{h.c.}$

c. $H_{\text{ind}}(f)\ G_d\ P_{\text{dir}}(d) + \text{h.c.}$

d. $H_{\text{ind}}(d)\ G_f\ P_{\text{dir}}(f) + \text{h.c.}$

As mentioned above, $a(b)$ and $c(d)$ are time-reversed partners of each other. We will now show that the contributions from these processes lead to a renormalisation of the inter-layer Kondo term. We suppress the Greens function since it is the same for all the terms:

$$G^{JK} = \frac{1}{\omega - |\epsilon_\alpha(\mathbf{q})|/2 + J_\alpha/4 + K_{\bar{\alpha}}/4 - J_\perp/4}~, \tag{3.31}$$

containing antiferromagnetic contributions from J_α and K_α and ferromagnetic contribution in $J_\perp$. The contribution from the pair is of the form

$$\begin{aligned}&\sum_\alpha \left[P_{\text{dir}}(\alpha)H_{\text{ind}}(\bar\alpha) + H_{\text{ind}}(\bar\alpha)P_{\text{dir}}(\alpha)\right]\\ &= \sum_{\alpha,a,b,\beta,\gamma,\delta}\frac{1}{4}J_\alpha(q,k')K_{\bar\alpha}(k,q)\left[\mathbf{S}^a_\alpha\sigma^a_{\beta,\gamma}c_{k',\gamma,\alpha}\mathbf{S}^b_{\bar\alpha}\sigma^b_{\delta,\beta}c^\dagger_{k,\delta,\alpha} + \mathbf{S}^b_{\bar\alpha}\sigma^b_{\delta,\beta}c^\dagger_{k,\delta,\alpha}\mathbf{S}^a_\alpha\sigma^a_{\beta,\gamma}c_{k',\gamma,\alpha}\right]\end{aligned} \tag{3.32}$$

The sum over β can be carried out:

$$\begin{aligned}&\sum_{\alpha,a,b,\beta,\gamma,\delta}\frac{1}{4}J_\alpha(q,k')K_{\bar\alpha}(k,q)\left[\mathbf{S}^a_\alpha\sigma^a_{\beta,\gamma}c_{k',\gamma,\alpha}\mathbf{S}^b_{\bar\alpha}\sigma^b_{\delta,\beta}c^\dagger_{k,\delta,\alpha} + \mathbf{S}^b_{\bar\alpha}\sigma^b_{\delta,\beta}c^\dagger_{k,\delta,\alpha}\mathbf{S}^a_\alpha\sigma^a_{\beta,\gamma}c_{k',\gamma,\alpha}\right]\\ &= \sum_{\alpha,a,b,\gamma,\delta}\frac{1}{4}J_\alpha(q,k')K_{\bar\alpha}(k,q)(\sigma^b\sigma^a)_{\delta,\gamma}\mathbf{S}^a_\alpha\mathbf{S}^b_{\bar\alpha}\left[c_{k',\gamma,\alpha}c^\dagger_{k,\delta,\alpha} + c^\dagger_{k,\delta,\alpha}c_{k',\gamma,\alpha}\right]\\ &= \sum_{\alpha,a,b,\gamma,\delta}\frac{1}{4}J_\alpha(q,k')K_{\bar\alpha}(k,q)(\sigma^b\sigma^a)_{\delta,\gamma}\mathbf{S}^a_\alpha\mathbf{S}^b_{\bar\alpha}\delta_{k,k'}\delta_{\delta,\gamma}\\ &= \sum_{\alpha,a,b}\frac{1}{4}J_\alpha(q,k')K_{\bar\alpha}(k,q)\text{Trace}(\sigma^b\sigma^a)\mathbf{S}^a_\alpha\mathbf{S}^b_{\bar\alpha}\delta_{k,k'}\ .\end{aligned} \tag{3.33}$$

Adding the contribution from the hermitian conjugate and restoring the Greens function, we get the total contribution as

$$\sum_\alpha \left[P_{\text{dir}}(\alpha)H_{\text{ind}}(\bar\alpha) + H_{\text{ind}}(\bar\alpha)P_{\text{dir}}(\alpha)\right] + \text{h.c.} = \frac{1}{2}\mathbf{S}_d\cdot\mathbf{S}_f\sum_{\alpha,q,k}\frac{J_\alpha(q,k)K_{\bar\alpha}(k,q)}{1/G^{JK} - J_\perp/4}\ . \tag{3.34}$$

The renormalisation of the inter-layer coupling arising from this process is

$$\frac{\Delta J_\perp}{\Delta D} = \int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\int_{IR} d\mathbf{k}\rho(\mathbf{k})\sum_{\alpha,q,k}\frac{J_\alpha(q,k)K_{\bar\alpha}(k,q)}{1/G^{JK} - J_\perp/4}\ . \tag{3.35}$$

1.3.4 Mixed processes

We now turn to the renormalisation arising from particle-hole mixed processes (designated as M_α above). There are four types of process:

$$\begin{aligned}P_1 &: \sum_\alpha M_{\text{dir}}(\alpha)GM_{\text{dir}}(\alpha)\\ P_2 &: \sum_\alpha M_{\text{ind}}(\alpha)GM_{\text{ind}}(\alpha)\\ P_3 &: \sum_\alpha\left[M_{\text{dir}}(\alpha)GM_{\text{ind}}(\bar\alpha) + \text{h.c.}\right]\\ P_4 &: \sum_\alpha\left[M_{\text{dir}}(\alpha)GM_{\text{ind}}(\alpha) + \text{h.c.}\right]\\ P_5 &: \sum_\alpha\left[M_{\text{dir}}(\alpha)GM_{\text{dir}}(\bar\alpha) + M_{\text{ind}}(\alpha)GM_{\text{ind}}(\bar\alpha)\right]\ .\end{aligned} \tag{3.36}$$

We first note that P_1 and P_2 are equivalent to the ones considered in the direct-direct and indirect-indirect subsections, specifically the renormalisation of $J_\perp$ in eq. 3.21 and its indirect counterpart, the only difference being that k will now range over the UV subspace instead of IR:

$$\frac{\Delta J_\perp}{\Delta D} = -\frac{1}{4}\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}') \int_{UV} d\mathbf{q}'\rho(\mathbf{q}') \sum_\alpha \Bigg[J_\alpha(\mathbf{q}',\mathbf{q})^2 \left(\frac{1}{1/G^J_\alpha + 3J/4} + \frac{1}{1/G^J_\alpha - J/4}\right) + K_\alpha(\mathbf{k},\mathbf{q})^2 \left(\frac{1}{1/G^K_{\bar\alpha} + 3J/4} + \frac{1}{1/G^K_{\bar\alpha} - J/4}\right)\Bigg] . \tag{3.37}$$

The other equations in the same subsection – the ones pertaining to the renormalisation of J_α or K_α are not relevant here because they require the participation of IR bath operators. Similarly, P_3 and P_4 are equivalent to the ones considered in the previous subsection (eq. 3.35), the only difference being that the dummy variable k in the previous subsection would now have to range only over the UV subspace:

$$\frac{\Delta J_\perp}{\Delta D} = \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \int_{UV} d\mathbf{q}'\rho(\mathbf{q}') \sum_\alpha \frac{J_\alpha(q,q')K_{\bar\alpha}(q',q)}{1/G^{JK}_\alpha - J_\perp/4} . \tag{3.38}$$

We finally evaluate the contribution from processes P_4 and P_5. In order to capture coherent scattering processes between the two layers arising from P_4, we project onto the following rotated basis of excited states:

$$\begin{aligned} |H\rangle_\pm &= \frac{1}{\sqrt{2}}\Big(|H(\mathbf{q})\rangle_f \pm |H(\mathbf{q})\rangle_d\Big) , \\ \mathcal{P}(\mathbf{q})_H &= |H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_- , \end{aligned} \tag{3.39}$$

where $|H(\mathbf{q})\rangle_\alpha$ is the state obtained upon exciting both states in the layer α: $|H(q)\rangle_\alpha = c^\dagger_{\mathbf{q},\alpha} c_{\bar{\mathbf{q}},\alpha} |L\rangle_f |L\rangle_d$, and $\mathcal{P}(\mathbf{q})_H$ projects onto this rotated excited basis. We focus on the spin-flip component $JS^+_f S^-_d$ of the Hamiltonian. There are two kinds of processes that renormalise this coupling. We first consider one that involves a spin-flip of the $d-$layer followed by a spin-flip of the $f-$layer:

$$\Delta H = \frac{1}{N^2}\sum_{\mathbf{q}\in\mathrm{UV}} \langle L|\frac{1}{2}J_f(\bar{\mathbf{q}},\mathbf{q}) S^+_f c^\dagger_{\mathbf{q}-\downarrow,f} c_{\mathbf{q}+\uparrow,f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H \frac{1}{2} J_d(\mathbf{q},\bar{\mathbf{q}}) S^-_d c^\dagger_{\mathbf{q}\uparrow,d} c_{\bar{\mathbf{q}}\downarrow,d} |L\rangle , \tag{3.40}$$

where $|L\rangle = |L\rangle_f |L\rangle_d$ is the initial configuration for both layers. G is the propagator for the excited state:

$$G = \frac{1}{\omega - H_D} , \tag{3.41}$$

where H_D is the diagonal part of the Hamiltonian corresponding to the excited(intermediate) state. The diagonal part consists of the kinetic energy and the Ising part of the Kondo interaction:

$$H_D = \sum_{\mathbf{q},\sigma\in\mathbf{q}_\pm,\alpha} \epsilon_\alpha(\mathbf{q})\tau_{\mathbf{q},\sigma} + \frac{1}{2}\sum_\alpha J_\alpha S^z_\alpha \sum_{\mathbf{q},\sigma} \sigma n_{\mathbf{q},\sigma,\alpha} + J_\perp \vec{S}_f\cdot\vec{S}_d . \tag{3.42}$$

Calculating the matrix elements of the propagator in the rotated basis gives

$$\begin{aligned}\mathcal{P}(\mathbf{q})_H G\mathcal{P}(\mathbf{q})_H =&\frac{1}{2}\left(\frac{1}{1/G^J_{ff}-J_\perp/4}+\frac{1}{1/G^J_{dd}-J_\perp/4}\right)\left(|H\rangle_+\langle H|_+ + |H\rangle_-\langle H|_-\right)\\ &+\frac{1}{2}\left(\frac{1}{1/G^J_{ff}-J_\perp/4}-\frac{1}{1/G^J_{dd}-J_\perp/4}\right)\left(|H\rangle_+\langle H|_- + |H\rangle_-\langle H|_+\right)\ ,\end{aligned} \tag{3.43}$$

where

$$G^J_{\alpha\alpha}(\mathbf{q}) = \frac{1}{\omega-\frac{1}{2}\left[\epsilon_\alpha(\mathbf{q})-\epsilon_\alpha(\bar{\mathbf{q}})\right]+\frac{1}{2}J_\alpha(\mathbf{q})+W_\alpha(\mathbf{q})}\ . \tag{3.44}$$

where $\bar{\epsilon}_\alpha(\mathbf{q})=\epsilon_f(\mathbf{q})-\epsilon_f(\bar{\mathbf{q}})$ is the total excitation energy for the pair of states $\mathbf{q},\bar{\mathbf{q}}$.

In eq. 3.40, since the first process is an excitation into a $d-$state and the second process is a de-excitation from an $f-$state, the expression can be simplified into

$$\begin{aligned}\langle L|S_f^+c^\dagger_{\mathbf{q}-\downarrow,f}c_{\mathbf{q}+\uparrow,f}\mathcal{P}(\mathbf{q})_H G\mathcal{P}(\mathbf{q})_H S_d^- c^\dagger_{\mathbf{q}+\uparrow,d}c_{\mathbf{q}-\downarrow,d}|L\rangle &= S_f^+S_d^-\langle H_f|\mathcal{P}(\mathbf{q})_H G\mathcal{P}(\mathbf{q})_H|H_d\rangle\\ &=\frac{1}{2}\left(\frac{1}{1/G^J_{ff}-J_\perp/4}+\frac{1}{1/G^J_{dd}-J_\perp/4}\right)S_f^+S_d^-\ .\end{aligned} \tag{3.45}$$

The net Hamiltonian renormalisation is finally a sum over the contribution from each mode:

$$\Delta H = \frac{1}{8N^2}\sum_{\mathbf{q}\in\mathrm{UV}} J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})\left(\frac{1}{1/G^J_{ff}-J_\perp/4}+\frac{1}{1/G^J_{dd}-J_\perp/4}\right)S_f^+S_d^-\ . \tag{3.46}$$

Converting the sums to integrals (assuming a density of states $\rho(\mathbf{q})$) gives the final expression

$$\frac{\Delta H}{\Delta D} = \frac{1}{8N}S_f^+S_d^-\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})\left(\frac{1}{1/G^J_{ff}(q)-J_\perp/4}+\frac{1}{1/G^J_{dd}(q)-J_\perp/4}\right)\ . \tag{3.47}$$

The time-reversed partner of this process can be obtained simply by exchanging the $d-$interaction with the $f-$interaction. Since the renormalisation itself is symmetric, the contribution is equal to what we obtained here. The total renormalisation in the coupling for the term $S_f^+S_d^-$ is therefore

$$\frac{\Delta J_\perp}{\Delta D} = \frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})\left(\frac{1}{1/G^J_{ff}(q)-J_\perp/4}+\frac{1}{1/G^J_{dd}(q)-J_\perp/4}\right)\ . \tag{3.48}$$

Taking into account P_5 by replacing couplings J_α with K_α and the Greens function $G^J_{\alpha\alpha}$ with

$$G^K_{\alpha\alpha}(\mathbf{q}) = \frac{1}{\omega-\frac{1}{2}\left[\epsilon_\alpha(\mathbf{q})-\epsilon_\alpha(\bar{\mathbf{q}})\right]+\frac{1}{2}K_{\bar{\alpha}}(\mathbf{q})+W_\alpha(\mathbf{q})}\ , \tag{3.49}$$

we have the total renormalisation

$$\begin{aligned}\frac{\Delta J_\perp}{\Delta D} = \frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\Bigg[&J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})\sum_\alpha\frac{1}{1/G^J_{\alpha\alpha}(\mathbf{q})-J_\perp/4}\\ &+K_f(\bar{\mathbf{q}},\mathbf{q})K_d(\mathbf{q},\bar{\mathbf{q}})\sum_\alpha\frac{1}{1/G^K_{\alpha\alpha}(\mathbf{q})-J_\perp/4}\Bigg]\ .\end{aligned} \tag{3.50}$$

1.3.5 Complete RG Equations

Inter-layer Kondo coupling $J_\perp$

$$\begin{aligned}\frac{\Delta J_\perp}{\Delta D} =& -\frac{1}{4}\int_{\text{UV}} dq\rho(q)\int_{\text{UV + IR}} dk\rho(k)\sum_\alpha\left[J_\alpha^2(q,k)\left(\frac{1}{1/G_\alpha^J + 3J_\perp/4} + \frac{1}{1/G_\alpha^J - J_\perp/4}\right)\right.\\ &\left. K_\alpha^2(q,k)\left(\frac{1}{1/G_\alpha^K + 3J_\perp/4} + \frac{1}{1/G_\alpha^K - J_\perp/4}\right)\right]\\ &+\int_{\text{UV}} dq\rho(q)\int_{\text{UV + IR}} dk\rho(k)\sum_\alpha\frac{J_\alpha(q,k)K_{\bar\alpha}(k,q)}{1/G_\alpha^{JK} - J_\perp/4}\\ &+\frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\left[J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})\sum_\alpha\frac{1}{1/G_{\alpha\alpha}^J(\mathbf{q}) - J_\perp/4}\right.\\ &\left. + K_f(\bar{\mathbf{q}},\mathbf{q})K_d(\mathbf{q},\bar{\mathbf{q}})\sum_\alpha\frac{1}{1/G_{\alpha\alpha}^K(\mathbf{q}) - J_\perp/4}\right]\end{aligned} \tag{3.51}$$

Direct Kondo coupling J_α

$$\begin{aligned}\frac{\Delta J_\alpha(k_1,k_2)}{\Delta D} =& -\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\left[J_\alpha(\mathbf{k}_1,\mathbf{q})J_\alpha(\mathbf{q},\mathbf{k}_2) + 4J_\alpha(\mathbf{q},\bar{\mathbf{q}})W_\alpha(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)\right]\\ &\left(\frac{1/4}{1/G_\alpha^J(\mathbf{q}) + 3J/4} + \frac{3/4}{1/G_\alpha^J(\mathbf{q}) - J/4}\right)\\ &+\frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\left[K_{\bar\alpha}(\mathbf{k}_1,\mathbf{q})K_{\bar\alpha}(\mathbf{q},\mathbf{k}_2) - 4K_{\bar\alpha}(\mathbf{q},\bar{\mathbf{q}})W_\alpha(\mathbf{k}_1,\mathbf{q},\mathbf{q},\mathbf{k}_2)\right]\\ &\left(\frac{1}{1/G_\alpha^K + 3J/4} + \frac{1}{1/G_\alpha^K - J/4}\right)\ .\end{aligned} \tag{3.52}$$

Indirect Kondo coupling K_α

$$\begin{aligned}\frac{\Delta K_\alpha(k_1,k_2)}{\Delta D} =& -\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\left[K_\alpha(\mathbf{k}_1,\mathbf{q})K_\alpha(\mathbf{q},\mathbf{k}_2) + 4K_\alpha(\mathbf{q},\bar{\mathbf{q}})W_{\bar\alpha}(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)\right]\\ &\left(\frac{1}{4}\frac{1}{1/G_\alpha^K(\mathbf{q}) + 3J/4} + \frac{3}{4}\frac{1}{1/G_\alpha^K(\mathbf{q}) - J/4}\right)\\ &+\frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\left[J_{\bar\alpha}(\mathbf{k}_1,\mathbf{q})J_{\bar\alpha}(\mathbf{q},\mathbf{k}_2) - 4J_{\bar\alpha}(\mathbf{q},\bar{\mathbf{q}})W_{\bar\alpha}(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)\right]\\ &\left(\frac{1}{1/G_{\bar\alpha}^J + 3J/4} + \frac{1}{1/G_{\bar\alpha}^J - J/4}\right)\ .\end{aligned} \tag{3.53}$$

Propagators

$$\begin{aligned}
\text{Intra-layer spin-flip excitation: } G^J_\alpha(\mathbf{q}) &= \frac{1}{\omega - |\epsilon_\alpha(\mathbf{q})|/2 + J_\alpha(\mathbf{q})/4 + W_\alpha(\mathbf{q})/2} ,\\
\text{Inter-layer spin-flip excitation: } G^K_\alpha(\mathbf{q}) &= \frac{1}{\omega - |\epsilon_{\bar\alpha}(\mathbf{q})|/2 + K_\alpha(\mathbf{q})/4 + W_{\bar\alpha}(\mathbf{q})/2} ,\\
\text{Both spins, one of the baths: } G^{JK}(\mathbf{q}) &= \frac{1}{\omega - |\epsilon_\alpha(\mathbf{q})|/2 + J_\alpha/4 + K_{\bar\alpha}/4 - J_\perp/4} ,\\
\text{Intra-layer spin-flip, two UV excitation: } G^J_{\alpha\alpha}(\mathbf{q}) &= \frac{1}{\omega - \frac{1}{2}\left[\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})\right] + \frac{1}{2}J_\alpha(\mathbf{q}) + W_\alpha(\mathbf{q})} ,\\
\text{Inter-layer spin-flip, two UV excitation: } G^K_{\alpha\alpha}(\mathbf{q}) &= \frac{1}{\omega - \frac{1}{2}\left[\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})\right] + \frac{1}{2}K_{\bar\alpha}(\mathbf{q}) + W_\alpha(\mathbf{q})} .
\end{aligned} \tag{3.54}$$

1.3.6 Simplified limit of $s-$wave scattering and $J_\alpha \ll D, K_\alpha \ll D, J_\perp \ll D$

We have

$$G^J_\alpha = G^K_\alpha = G^{JK} = G_1, \;\; G^J_{\alpha\alpha} = G^K_{\alpha\alpha} = G_2 \tag{3.55}$$

$$\begin{aligned}
\frac{\Delta J_\perp}{\Delta D} &= -\frac{1}{2}\int_{\text{UV}} d\mathbf{q}\rho(\mathbf{q}) \sum_\alpha \left[(J_\alpha + K_{\bar\alpha})^2 G_1 + G_2(J_f J_d + K_f K_d)\right]\\
\frac{\Delta J_\alpha}{\Delta D} &= -\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[J_\alpha^2 + 4J_\alpha W_\alpha\right] G_1 + \frac{1}{2}\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) G_1 \left[K_{\bar\alpha}^2 - 4K_{\bar\alpha}W_\alpha\right] ,\\
\frac{\Delta K_\alpha}{\Delta D} &= -\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[K_\alpha^2 + 4K_\alpha W_\alpha\right] G_1 + \frac{1}{2}\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) G_1 \left[J_{\bar\alpha}^2 - 4J_{\bar\alpha}W_\alpha\right] .
\end{aligned} \tag{3.56}$$

In ΔK_α, if K_α starts out as vanishingly small, the second term will dominate ($J \gg K$), so K_α will start off ferromagnetic (G_1 is negative). This ferromagnetic K_α can frustrate J_α through the second term in ΔJ_α. Moreover, it competes with J_α in trying to renormalise $J_\perp$ (in the first term at least).

1.4 Numerical Implementation

Since the bath interaction doesn't renormalise, it's functional form remains unchanged and can be substituted into the RG equations. In fact, we can write the expressions in a compact matrix form for efficient computation:

$$\frac{\Delta J_\alpha}{\Delta D} = -J_\alpha \mathcal{G}_\alpha J_\alpha^\dagger - 4\mathcal{W}_\alpha \mathrm{Tr}\left[\mathcal{G}'_\alpha\right] \ , \tag{4.1}$$

where all objects are matrices. J_α is the Kondo coupling matrix. $\mathcal{G}_\alpha$ is a diagonal matrix with matrix elements

$$\mathcal{G}_\alpha[q,q] = \begin{cases} \rho(q)\left(\frac{1}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})+\mu/2+3J/4} + \frac{3}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})+\mu/2-J/4}\right), \ \mathbf{q}\in \mathrm{PS}\ , \\ \rho(q)\left(\frac{1}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})-\mu/2+3J/4} + \frac{3}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})-\mu/2-J/4}\right), \ \mathbf{q}\in \mathrm{HS}\ , \end{cases} \tag{4.2}$$

if q is in the UV subspace and zero otherwise. $\mathcal{W}_\alpha$ is the bath interaction matrix whose matrix elements are given by $\mathcal{W}_\alpha[k,k'] = \frac{W}{2}\left[\cos(k_x - k'_x) + \cos(k_y - k'_y)\right]$. Finally, $\mathcal{G}'_\alpha$ is another diagonal matrix whole matrix elements are defined as $\mathcal{G}'_\alpha[q,q] = \mathcal{G}[q,q]J_\alpha[q,\bar{q}]$.

The RG equation for the inter-layer coupling J can be written as

$$\frac{\Delta J}{\Delta D} = \frac{1}{2}\mathbf{J}\cdot\mathbf{G}\ , \tag{4.3}$$

where $\mathbf{J}$ and $\mathbf{G}$ are vectors defined by the elements $\mathbf{J}[q] = J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})$ and

$$\mathbf{G}[q] = \rho(\mathbf{q})\left[\frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2}J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4}J} + \frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2}J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4}J}\right]. \tag{4.4}$$

1.5 Renormalisation Group Flows of Kondo Couplings: RG Phase Diagram

We first analyse the model by obtaining unitary RG flows of three kinds of Kondo couplings. Fig. 1.1 shows the fixed point values of the couplings under renormalisation to a low-energy theory. We view these values as a function of the $f-$layer bath interaction W_f and inter-layer Kondo coupling $J_\perp$. In each panel, the right (left) colorbar quantifies the background (marker) color of the plot. The background displays the fixed point value of each coupling whereas the marker color tracks whether the couplings is relevant (1), intermediate coupling (0.5) or irrelevant (0).

The Kondo coupling J_d (middle panel) of the $d-$layer remains relevant for the entirety of the chosen parameter regime. The Kondo coupling J_f (left panel) of the $f-$layer becomes RG-irrelevant for larger values of $|W_f|$; this leads to an in-plane Mott transition of the kind observed in our earlier work, where the local Fermi liquid excitations of the $f-$impurity give way to a pseudogapped non-Fermi liquid Mott metal phase which ultimately destabilises into a local moment phase. The RG flow of J_f does not appear to be significantly affected by the variation of $J_\perp$. Concomitant with

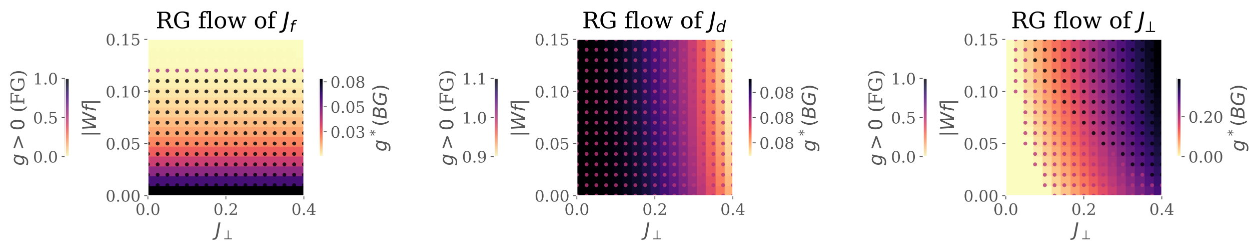

Figure 1.1: *RG phase diagrams for various Kondo couplings: $f-$layer (left), $d-$layer (middle) and inter-layer (right). For each plot, the background color (quantified by the right colorbar) tracks the fixed point value of the coupling as a function of the inter-layer couplings $J_\perp$ and the $f-$layer bath interaction $|W_f|$. The marker color complements this by tracking whether the coupling is relevant, intermediate-coupling or irrelevant in the RG sense, by taking the values 1, $1/2$ or 0 respectively.*

the irrelevance of J_f under increasing $|W_f|$, the inter-plane coupling $J_\perp$ (right panel) becomes RG-relevant, as seen from the darkening markers as we go to higher $|W_f|$.

These simulations motivate an RG-based phase diagram with the following regimes: (i) weak $J_\perp$, weak $|W_f|$: decoupled metallic layers, each layer participating in transport independently, (ii) strong $J_\perp$, strong $|W_f|$: Kondo insulator regime, where the local moments of the $f-$layer hybridise with the itinerant electrons of the $d-$layer to form a band insulator with the chemical potential situated in the hybridisation gap, and (iii) weak $J_\perp$, strong $|W_f|$: the so-called RKKY regime, where the $f-$layer forms local moments which decouple from the $d-$layer and are then free to order magnetically, and (iv) strong $J_\perp$, weak $|W_f|$: this is similar to the Kondo insulator regime, except here the $f-$electrons remain itinerant and it is they who hybridise with the $d-$electrons to again form two bands separated by a hybridisation gap.

1.6 Results at half-filling

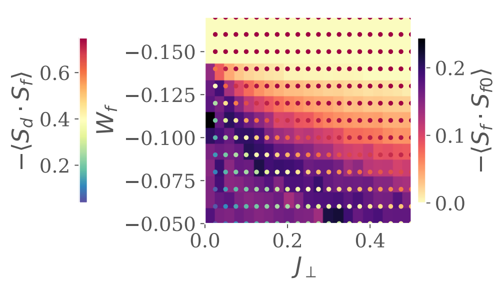

Figure 1.2: *Left: Real-space spin correlations for the auxiliary model. Background colour (right colourbar) represents that between the $f-$impurity and its nearest-neighbour sites, while the marker colour (left colourbar) represents that between the $f-$impurity and the $d-$impurity. While the former shrinks as $J_\perp$ or $|W_f|$ is increased, the latter grows. This marks a transfer of entanglement from intra-plane channel into inter-plane channel. Right:*

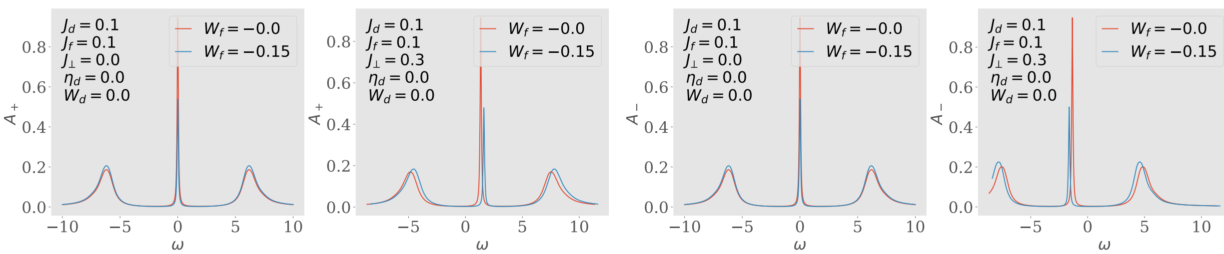

Figure 1.3: *Spectral functions of the tiled model, for excitations in the bonding (two on the left) and antibonding (two on the right) basis, defined by the Greens function* $G_{k,\pm} = G(k_d \pm k_f; \omega)$. *There is no hybridisation for* $J_\perp = 0$ *(first and third plots), and the only effect of* W_f *is to shrink the height of the central resonance. Introducing a non-zero* $J_\perp$ *splits the central resonance into an upper band (second from left) and a lower band (first from right). This leads to a Kondo insulator, where* $\omega = 0$ *resides inside the hybridisation gap.*

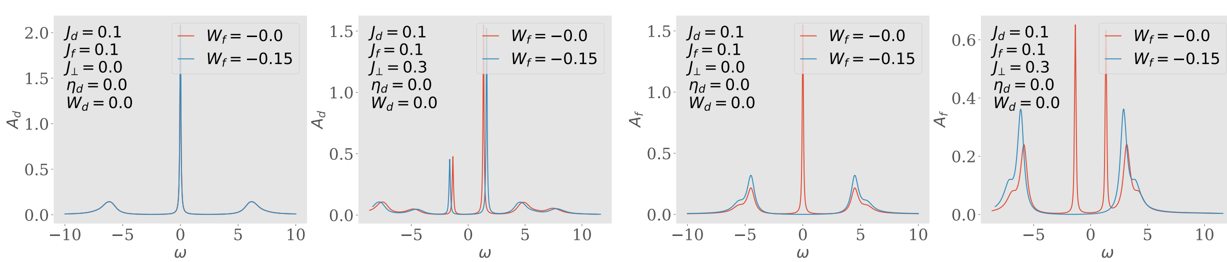

Figure 1.4: *Spectral functions* A_d *and* A_f *of the auxiliary model, for excitations in the* $d-$ *and* $f-$*layers respectively. Both show a central Kondo resonance for* $J_\perp = 0$ *which splits into two bands upon adding a non-zero* $J_\perp$. *Increasing* $|W_f|$ *to large values gaps out the local Fermi liquid excitations of* A_f.

Bibliography

[1] Abhirup Mukherjee, N S Vidhyadhiraja, A Taraphder, and Siddhartha Lal. Kondo frustration via charge fluctuations: a route to mott localisation. *New Journal of Physics*, 25(11):113011, nov 2023.